

LAPORAN PRAKTIKUM
Aplikasi Berbasis Platform

MODUL 12-13



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PROGRAM STUDI S1 INFORMATIKA

FAKULTAS INFORMATIKA

TELKOM UNIVERSITY

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Source code

main.dart

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
import 'providers/counter_provider.dart';
import 'services/notification_service.dart';
import 'screens/home_screen.dart';

void main() async {
  // Memastikan binding Flutter siap sebelum inisialisasi service
  WidgetsFlutterBinding.ensureInitialized();

  // Inisialisasi fungsi notifikasi lokal
  await NotificationService.init();

  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return ChangeNotifierProvider(
      create: (context) => CounterProvider(),
      child: MaterialApp(
        title: 'Flutter Demo',
        debugShowCheckedModeBanner: false,
        theme: ThemeData(
          useMaterial3: true,
          colorScheme: ColorScheme.fromSeed(seedColor: Colors.blue),
        ),
        home: const HomeScreen(),
      ),
    );
  }
}
```

lib/services/notification_service.dart

```
import 'package:flutter_local_notifications/flutter_local_notifications.dart';

class NotificationService {
  static final FlutterLocalNotificationsPlugin _notificationsPlugin =
    FlutterLocalNotificationsPlugin();

  // Inisialisasi Notifikasi
  static Future<void> init() async {
    // Pengaturan untuk Android (menggunakan ikon bawaan aplikasi)
    const AndroidInitializationSettings initializationSettingsAndroid =
      AndroidInitializationSettings('@mipmap/ic_launcher');

    const InitializationSettings initializationSettings = InitializationSettings(
      android: initializationSettingsAndroid,
    );

    await _notificationsPlugin.initialize(initializationSettings);

    // Request izin notifikasi untuk Android 13+ (API 33+)
    _notificationsPlugin
      .resolvePlatformSpecificImplementation<
        AndroidFlutterLocalNotificationsPlugin>()
      ?.requestNotificationsPermission();
  }

  // Fungsi untuk menampilkan notifikasi
  static Future<void> showNotification(int counterValue) async {
    const AndroidNotificationDetails androidPlatformChannelSpecifics =
      AndroidNotificationDetails(
        'counter_channel_id', // ID Channel
        'Counter Updates',    // Nama Channel
        channelDescription: 'Notifikasi saat nilai counter berubah',
        importance: Importance.max,
        priority: Priority.high,
        showWhen: false,
      );

    const NotificationDetails platformChannelSpecifics = NotificationDetails(
      android: androidPlatformChannelSpecifics,
    );

    await _notificationsPlugin.show(
```

```
    0, // Notification ID
    'Counter Update', // Judul Notifikasi
    'Nilai counter saat ini: $counterValue', // Pesan Notifikasi
    platformChannelSpecifics,
  );
}
```

lib/providers/counter_provider.dart

```
import 'package:flutter/material.dart';
import '../services/notification_service.dart';

class CounterProvider with ChangeNotifier {
  int _counter = 0;

  // Getter untuk mengambil nilai counter
  int get counter => _counter;

  // Fungsi untuk menambah counter dan memicu notifikasi
  void increment() {
    _counter++;
    notifyListeners(); // Memberitahu UI untuk merender ulang

    // Panggil fungsi notifikasi
    NotificationService.showNotification(_counter);
  }
}
```

lib/screens/home_screen.dart

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
import '../providers/counter_provider.dart';

class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Counter & Notification'),
        backgroundColor: Colors.blue,
        foregroundColor: Colors.white,
      ),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: <Widget>[
            const Text(
              'Nilai counter saat ini:',
              style: TextStyle(fontSize: 18, color: Colors.grey),
            ),
            const SizedBox(height: 10),
            // Menggunakan Consumer untuk mendengarkan perubahan pada CounterProvider
            Consumer<CounterProvider>(
              builder: (context, counterProvider, child) {
                return Text(
                  '${counterProvider.counter}',
                  style: const TextStyle(
                    fontSize: 64,
                    fontWeight: FontWeight.bold,
                    color: Colors.blue,
                  ),
                );
              },
            ),
          ],
        ),
      ),
    );
  }
}
```

Output

```
floatingActionButton: FloatingActionButton(  
  onPressed: () {  
    // Memanggil fungsi increment tanpa mendengarkan perubahan di sini (listen: false)  
    Provider.of<CounterProvider>(context, listen: false).increment();  
  },  
  tooltip: 'Tambah',  
  backgroundColor: Colors.blue,  
  child: const Icon(Icons.add, color: Colors.white),  
)  
);  
}
```

Nilai counter saat ini:

14



Aplikasi ini untuk melakukan counting ketika tombol ditekan.